

S HECATE STRAIT BUOY 46185, CanadaWMO: **996530**Lat: **52.420N**Lon: **129.790W**Elev: **3**StdP: **101.29**Time Zone: **-9.00 (W09)**Period: **01-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.1	0.1	N/A	N/A	N/A	N/A	N/A	N/A	19.0	7.1	17.9	7.6	8.5	20	0.816

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 1.6	(c) 17.2	(d) N/A	(e) 16.5	(f) N/A	(g) 15.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.1	(p) 310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	16.2	14.4	-2.9	18.4	3.1	2.5	-5.1	20.2	-6.9	21.7	-8.6	23.1	-10.8	24.9		
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	6.5	6.1	6.1	7.4	9.3	11.7	13.6	14.6	13.3	11.0	8.2	6.7
	DBStd	3.53	2.51	2.15	2.18	1.37	1.26	1.44	1.30	1.34	1.04	1.42	2.33	2.29
	HDD10.0	615	109	110	120	78	27	2	0	0	0	7	59	103
	HDD18.3	3201	367	343	378	328	279	198	147	116	149	229	305	361
	CDD10.0	457	0	0	0	1	7	54	111	143	101	37	4	1
	CDD18.3	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.2	8.9	7.9	8.1	6.9	5.8	5.8	5.7	5.5	6.4	7.6	8.7	9.0
Precipitation	PrecAvg	2965	351	250	268	212	139	120	101	135	205	375	423	387
	PrecMax	3703	526	430	524	407	284	205	293	284	380	604	607	654
	PrecMin	2385	170	109	102	87	46	43	22	33	32	169	247	159
	PrecStd	351	89	83	80	56	51	38	60	57	85	112	99	104
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	10.2	10.2	10.9	12.5	16.7	17.1	18.7	16.6	14.4	12.6	12.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	10.0	9.6	9.6	10.0	11.7	15.3	16.4	17.9	15.8	13.7	11.7	10.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	9.5	9.0	9.0	9.6	11.3	14.4	15.9	17.1	15.3	13.2	11.2	10.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	9.1	8.5	8.4	9.1	11.0	13.7	15.4	16.3	14.9	12.7	10.7	9.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.2	2.0	2.2	1.8	1.5	1.5	1.5	1.6	1.5	1.7	2.2	2.3
		MCDBR	2.2	2.0	2.2	2.0	1.8	2.0	1.9	2.2	1.7	1.9	2.2	2.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.317	0.323	0.349	0.385	0.383	0.374	0.370	0.373	0.364	0.347	0.324	0.309	
	taud	2.414	2.497	2.434	2.313	2.381	2.448	2.512	2.505	2.500	2.461	2.377	2.358	
	Ebn at Noon	682	802	844	849	869	879	877	855	820	755	655	613	
	Edh at Noon	58	71	91	115	113	107	99	94	85	72	59	51	
All-Sky Solar Radiation	RadAvg	0.61	1.47	2.46	3.90	5.04	5.28	5.10	4.35	2.99	1.65	0.78	0.51	
	RadStd	0.08	0.26	0.34	0.25	0.49	0.48	0.62	0.46	0.33	0.24	0.11	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page